System Flowchart

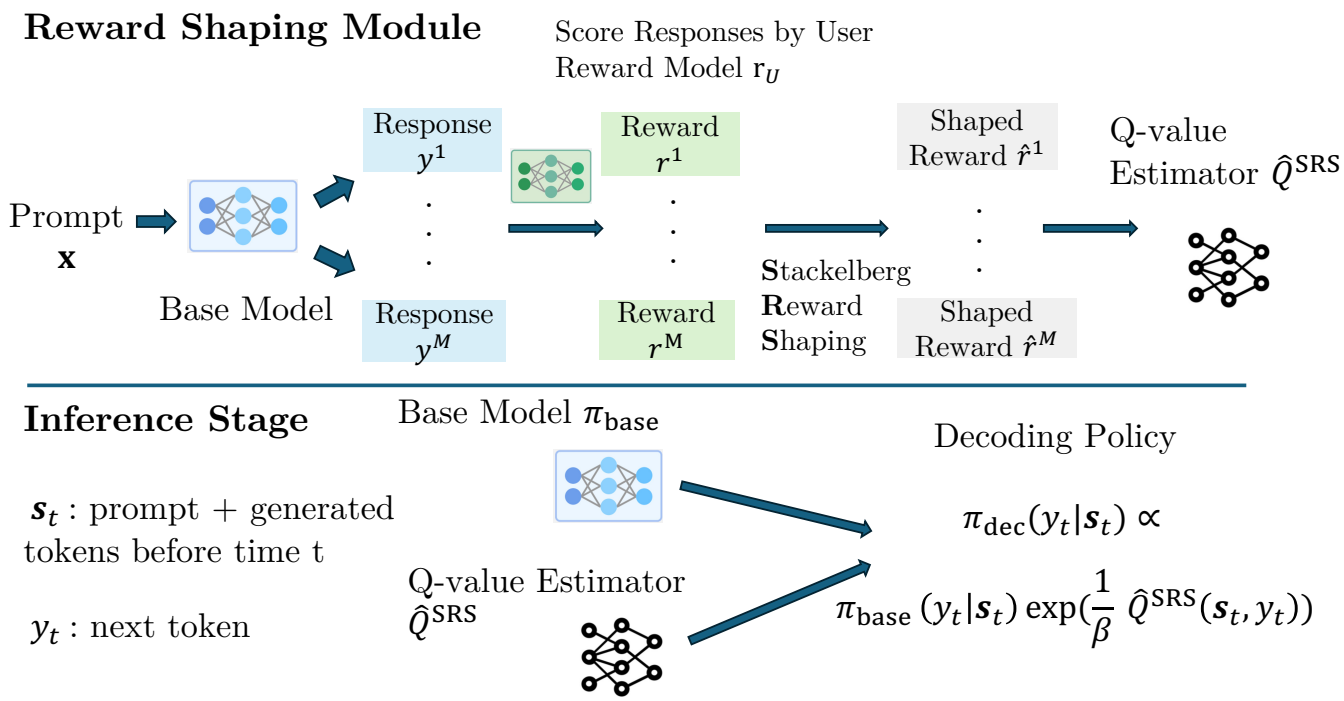

Figure 1: Overall framework of Stackelberg Reward Shaping. We first sample multiple responses from the base model and evaluate them using the user reward model r_U, followed by Stackelberg reward shaping to maximize user utility. This process produces a Q-value Estimator $\hat{Q}^{\text{SRS}}$, which captures the expected reward of completed responses under the shaped reward. During inference, $\hat{Q}^{\text{SRS}}$ is used to reweight the base model's token probabilities, promoting tokens aligned with user preferences.